

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

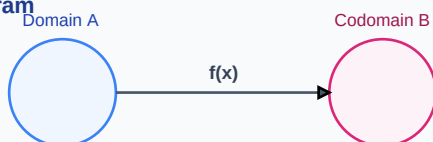
CHAPTER 1: RELATIONS AND FUNCTIONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Relations and Functions

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Cartesian Product	For non-empty sets A and B, $A \times B$ is the set of all ordered pairs (a, b) such that a belongs to A and b belongs to B.
Relation	A relation R from A to B is a subset of the Cartesian Product $A \times B$, satisfying specific conditions.
Function	A relation f from A to B is a function if for each element a in A, there exists a unique element b in B such that (a, b) belongs to f.
Domain and Range	For $f: A \rightarrow B$, A is the domain, B is the co-domain, and the set of all image values is the range of f.

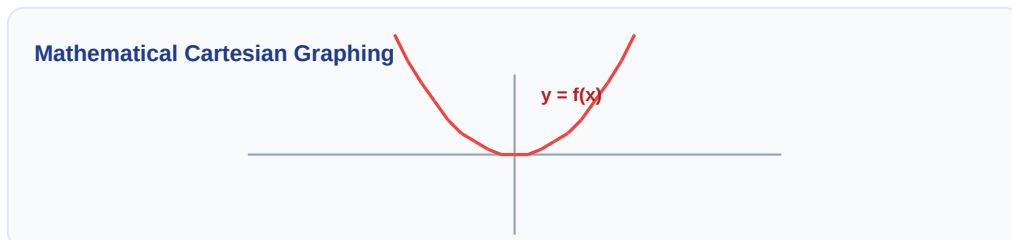
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Number of elements: $n(A \times B) = n(A) * n(B)$
- Identity Function: $f(x) = x$ for all x in A
- Constant Function: $f(x) = c$ (a constant) for all x in A
- Linear Function: $f(x) = mx + c$ where m is not zero
- Composition of Functions: $(f \circ g)(x) = f(g(x))$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: If $A = \{1, 3, 5\}$ and $B = \{2, 3\}$, find $A \times B$ and $B \times A$.

Step-by-step Solution:

$$A \times B = \{(1, 2), (1, 3), (3, 2), (3, 3), (5, 2), (5, 3)\}.$$

$$B \times A = \{(2, 1), (2, 3), (2, 5), (3, 1), (3, 3), (3, 5)\}.$$

Note: $A \times B$ is not equal to $B \times A$ since the ordered pairs differ.

Example 2: Given $f(x) = 2x + 1$ and $g(x) = x^2 - 2$, find $(f \circ g)(x)$.

Step-by-step Solution:

$$(f \circ g)(x) = f(g(x)) = f(x^2 - 2).$$

$$\text{Substitute } (x^2 - 2) \text{ in } f: 2(x^2 - 2) + 1 = 2x^2 - 4 + 1 = 2x^2 - 3.$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

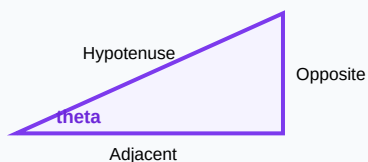
HOTS Challenge 1: Let $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$. Can a relation $R = \{(1, 5), (2, 6), (3, 7), (1, 6)\}$ be a function? Justify.

Analytical Solution Strategy:

No. In a function, each element in the domain must map to a unique element in the co-domain.

Here, the element 1 in domain A maps to two different images, 5 and 6 in co-domain B. Thus, R is not a function.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find $n(A \times B)$ if $n(A) = 5$ and $n(B) = 3$.

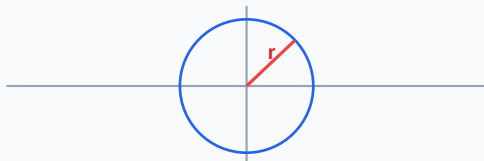
Q2 (2-Mark): Determine if the relation $R = \{(1,2), (2,3), (3,4)\}$ on $A = \{1,2,3,4\}$ is a function.

Q3 (2-Mark): If $f(x) = 3x - 2$, find $f(2)$ and $f(-1)$.

Q4 (5-Mark): If $g(x) = x + 3$ and $h(x) = 2x$, show that composition is associative: $g \circ (h \circ f) = (g \circ h) \circ f$.

Q5 (5-Mark): A function is defined by $f(x) = x^2 - x$. Solve $f(x) = 6$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Database Query Mapping

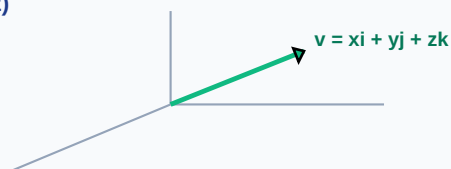
Cartesian products form the basis of SQL JOIN operations in databases, matching records of one table with another.

Application Case: Control Engineering

Functions model inputs and outputs in feedback control systems, predicting system stability based on mathematical transfers.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

